Proceedings of the 2018
26th International Conference on Nuclear Engineering
ICONE26
July 22-26, 2018, London, England

ICONE26-82220

THE CHARACTERISTICS STUDY OF HELIUM-XENON MIXTURE IN CLOSED BRAYTON CYCLE FOR SPACE NUCLEAR REACTOR POWER

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ABSTRACT

Differing from the adoption of helium as working fluid of closed Brayton cycle (CBC) for terrestrial high temperature gas cooled reactor (HTGR) power plants, helium-xenon mixture with a proper molar weight was recommended as working fluid for space nuclear reactor power with CBC conversion. It is essential to figure out how the component of helium-xenon mixture affects the net system efficiency, in order to provide reference for the selection of appropriate cycle working fluid. After a discussion of the physical properties of different helium-xenon mixtures, the related physical properties are studied to analyze their affection on the key parameters of CBC, including adiabatic coefficient, recuperator effectiveness and normalized pressure loss coefficient. Then the comprehensive thermodynamics of CBC net system efficiency is studied in detail considering different helium-xenon mixtures. The physical properties study reveals that at 0.7 MPa and 400 K, the adiabatic coefficient of helium-xenon mixture increases with increased molar weight, from 0.400 (pure helium) to 0.414 (pure xenon), while recuperator effectiveness firstly increases and then decreases with the increase of molar weight, and the normalized pressure loss coefficient increases monotonically with molar weight increases. The thermodynamic analysis results show that the adiabatic coefficient has less effect on the net system efficiency, while the net system efficiency increases with increased recuperator effectiveness, and the net system efficiency decreases with normalized pressure loss coefficient increases. Finally, the mixture of helium-8.6% xenon was adopted as working fluid, instead of pure helium, for ensuring less turbine mechanicals (turbine and compressor) stages, and resulting maximum recuperator effectiveness. At the given cold / hot side temperature of 400 / 1300 K, the net system efficiency can reach 29.18% theoretically.

1. INTRODUCTION

In space exploration and construction of outer planet surface base, the demand for power supply is inseparable. Due to the advantages of small volume, light weight and long-time running[1], space nuclear reactor power (SNRP) has broad application prospect, in compared with chemical power, solar energy power and radioisotope power. After deployed 34 times in Earth orbits[2], several technology options for SNRP has been studied, such as gas cooled reactor with closed Brayton cycle (CBC) engine[3], liquid metal cooled reactor with Rankine cycle[4] or free piston Stirling engine[5], heat pipe cooled reactor with alkali metal thermal-to-electric conversion (AMTEC)[6]. Among these technology options, gas cooled reactor with CBC is the best candidate for the SNRP, which has high electric power level up to megawatt, due to its high net system efficiency and high specific power[7].

As for the terrestrial high temperature gas cooled reactor (HTGR) with CBC, helium is generally used as the coolant of HTGR as well as the working fluid of CBC. However, there has some different considerations for SNRP. In the design phase of SNRP, the weight and size of CBC, along as the series of turbine machinery, are important technical parameters. Helium-xenon mixture with proper molar weight was recommended as the working fluid of CBC, as well as coolant of HTGR for SNRP, in order to reduce the weight and size of CBC[8, 9], and reduce the series of turbine machinery[10]. Figure 1 display the schematic diagram of the coolant loop and major components of a SNRP with a CBC engine.

This paper begins with the discussion of the physical properties of helium-xenon mixture change with molar weight. Then the related physical properties are studied to analyze their affection on the key parameters of CBC. The last part is the comprehensive thermodynamics analysis of CBC, with different

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components of helium-xenon mixture. The objective of this paper is to provide technical guidance for the design of SNRP.

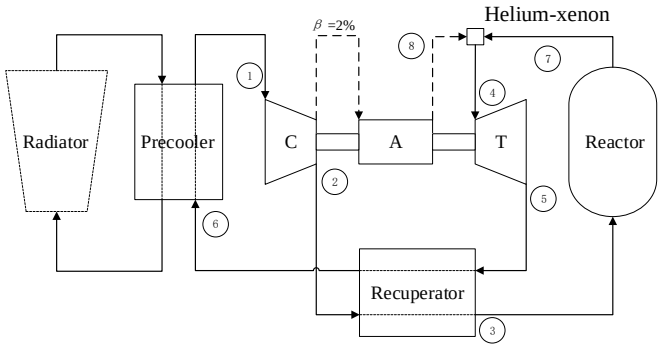


Fig. 1 Schematic diagram of the SNRP with CBC engine (A: alternator, C: compressor, T: turbine)

2. PHYSICAL PROPERTIES

Chapman-Enskog theory is widely used in the physical properties predication of dilute gases. However, after the addition of the high density gas, such as neon, argon, krypton, and xenon, to helium, the predicted properties based on Chapman-Enskong have great discrepancy in compared with experimental measurements. Therefore, Tournier et al[11] used the law of corresponding states to correct Chapman-Enskog approach and proposed semi-empirical correlations for accurately predicting the physical properties of noble gases and their binary mixtures. The application scope of these correlations is: pressure $P<20$MPa , and temperature T<1400K . According to these semi-empirical correlations, the physical properties of helium-xenon mixture can be calculated by the following formula:

$$P = RT\hat{\rho}Z = RT\hat{\rho}\left(1+B\hat{\rho}+C\hat{\rho}^2\right)$$
$$\rho = M\hat{\rho}, \quad M = x_1M_1 + x_2M_2 \tag{1}$$

$$\mu(T,P) = \mu^o(T) + \left(1+\frac{1}{2.3}\right)\mu^* \cdot \Psi_\mu\left(\frac{0.291\times V^*}{M}\rho\right) \tag{2}$$

$$\lambda(T,P) = \lambda^o(T) + \left(1+\frac{1}{2.94}\right)\lambda^* \cdot \Psi_\lambda\left(\frac{0.291\times V^*}{M}\rho\right) \tag{3}$$

$$\hat{C}_p = \hat{C}_p^o + \hat{\rho}R\left[\left(B - T\frac{dB}{dT} - T^2\frac{d^2B}{dT^2}\right) + \hat{\rho}\left(C - \frac{T^2}{2}\frac{d^2C}{dT^2}\right)\right] + RT\left[\left(B - T\frac{dB}{dT}\right) + \hat{\rho}\left(2C - T\frac{dC}{dT}\right)\right]\cdot\left[\frac{\partial\hat{\rho}}{\partial T}\right]_p \tag{4-1}$$

$$\hat{C}_v = \hat{C}_v^o - \hat{\rho}RT\left[\left(2\frac{dB}{dT} + T\frac{d^2B}{dT^2}\right) + \hat{\rho}\left(\frac{dC}{dT} + \frac{T}{2}\frac{d^2T}{dT^2}\right)\right] \tag{4-2}$$

$$\hat{C}_p^o = \frac{5}{2}R, \quad \hat{C}_v^o = \frac{3}{2}R \tag{4-3}$$

$$\left[\frac{\partial\hat{\rho}}{\partial T}\right]_p = -\frac{\left(\hat{\rho}+B\hat{\rho}^2+C\hat{\rho}^3\right)/T + \frac{dB}{dT}\hat{\rho}^2 + \frac{dC}{dT}\hat{\rho}^3}{1+2B\hat{\rho}+3C\hat{\rho}^2} \tag{4-4}$$

$$C_p = M\hat{C}_p, \quad C_v = M\hat{C}_v \tag{4-5}$$

where, ρ , μ , λ , C_P and C_V are helium-xenon mixture's density, dynamic viscosity, thermal conductivity, specific heat capacity at constant pressure and specific heat capacity at constant volume respectively. $\hat{\rho}$, $\hat{C}_P$ and $\hat{C}_V$ are helium-xenon mixture's molar density, molar specific heat capacity at constant pressure and molar specific heat capacity at constant volume respectively. M is the molar weight of helium-xenon mixture, and M_1 、 M_2 are the molar weight of pure helium and pure xenon respectively. x_1 is the molar fraction of helium, while x_2 is the molar fraction of xenon, x_1+x_2=1 . R is the perfect gas constant, 8.31441 J/(mole·K).

In equation (1), B is the second virial coefficient of helium-xenon mixture, and C is the third virial coefficient. In equation (2), the first term of right side is dynamic viscosity calculated by dilute gas theory, and the second term is the correction term based on the law of corresponding states. In equation (3), the first term of right side is thermal conductivity calculated by dilute gas theory, and the second term is the correction term based on the law of corresponding states. The detailed calculation formula of these parameters can be seen in reference [11].

Base on the correlations above, Fortran 95 was used to calculated the physical properties of helium-xenon mixture. Fig. 2 shows helium-xenon's compression factor Z , specific heat capacity at constant pressure C_p, dynamic viscosity μ , and thermal conductivity λ , change with molar weight M respectively, at pressure P=0.7 MPa and temperature T=400 K and 1300 K.

The first part of Fig. 2 shows that at low temperature (400 K), the compression factor of pure xenon is less than 0.99, which indicates that $P-V-T$ state equation for ideal gas is not appropriate to pure xenon, as well as helium-xenon mixture. According to the second part of Fig. 2, one can found that the specific heat capacity at constant pressure of helium-xenon mixture decreases with the increase of molar weight, and the decreasing rate is reduced with molar weight increases. Temperature has no effect on specific heat capacity at constant pressure of helium-xenon mixture. The last two parts of Fig. 2 show that the dynamic viscosity increases with molar weight increases, while thermal conductivity decreases with molar weight increases, and both of them increase with temperature increases.

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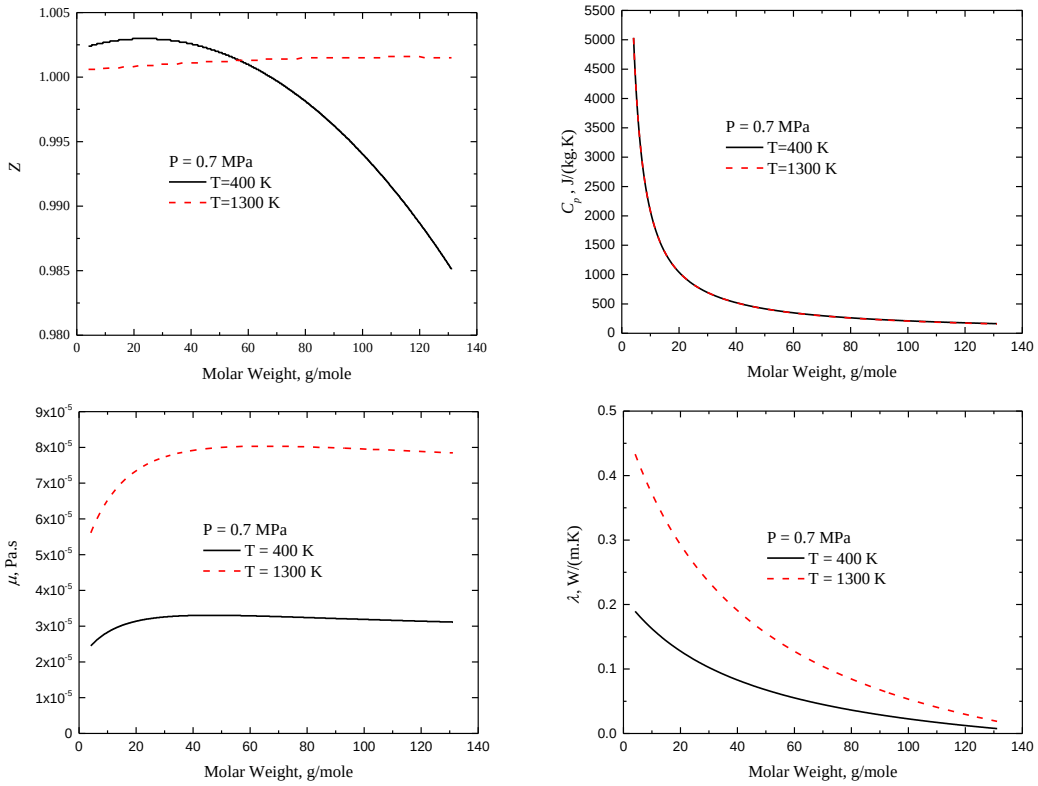


Fig. 2 Physical properties change with molar weight

3. THERMODYNAMIC MODELS

Before comprehensive thermodynamic analysis of CBC, it is necessary to understand the thermodynamic models, and to figure out the key parameters of CBC that affected by composition of helium-xenon mixture (molar weight). As seen in Fig. 1, six components are taken into consideration to establish the thermodynamic model of CBC.

3.1 Turbine

The expansion ratio π is defined as:

$$\pi = \frac{P_4}{P_5} \tag{5}$$

where P_4 and P_5 are the pressure of working fluid importing and exporting turbine respectively.

The ratio of pressure loss from 2⟶4 to the maximum pressure P_2, is represented as ξ_1. The ratio of pressure loss from 5⟶1 to the maximum pressure P_2, is represented as ξ_2. Therefore, P_4 and P_5 can be written as:

$$P_4 = P_2 - P_2 \cdot \xi_1 \tag{6}$$

$$P_5 = P_1 + P_2 \cdot \xi_2 \tag{7}$$

The compressor pressure ratio γ is defined as:

$$\gamma = \frac{P_2}{P_1} \tag{8}$$

According to equation (5) to (8), expansion ratio can be written as:

$$\pi = \frac{(1-\xi_1)\gamma}{1+\gamma\xi_2} \tag{9}$$

As for the ideal adiabatic expansion process, the ideal temperature of turbine outlet T_{5s} is expressed as:

$$T_{5s} = T_4 \cdot \pi^{-\varphi} \tag{10}$$

where $\varphi = (k\text{-}1)/k$ is adiabatic coefficient, and $k = C_p / C_v$ is the specific heat ration of working fluid. According to the expression of turbine adiabatic efficiency

$$\eta_T = \frac{T_4 - T_5}{T_4 - T_{5s}} \tag{11}$$

the temperature of turbine outlet T_5 is expressed as a function of the temperature of turbine inlet T_4, as:

$$T_5 = T_4\left(1 - \eta_T + \eta_T \pi^{-\varphi}\right) \tag{12}$$

The mechanical power developed in the turbine can be expressed in terms of the molar flow rate, molar specific heat, and the turbine inlet and outlet temperature as:

$$W_T = \dot{N}\hat{C}_{P,4-5}(T_4 - T_5) \tag{13}$$

Bring equation (12) to (13), one can get

$$W_T = \eta_T \dot{N}\hat{C}_{P,4-5}T_4(1-\pi^{-\varphi}) \tag{14}$$

3.2 Compressor

Similar to the turbine expansion process and according to the ideal adiabatic compression process, the ideal temperature

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of compressor outlet T_{2s} can be written as:

$$T_{2s} = T_1 \cdot \gamma^{\varphi} \tag{15}$$

According to the expression of compressor adiabatic efficiency

$$\eta_C = \frac{T_{2s} - T_1}{T_2 - T_1} \tag{16}$$

The temperature of compressor outlet T_2 is expressed as a function of the temperature of compressor inlet T_1, as:

$$T_2 = T_1\left[1 + \eta_C^{-1}(\gamma^{\varphi} - 1)\right] \tag{17}$$

The mechanical power used in the compressor W_C can be expressed in the terms of the molar flow rate, molar specific heat, and the compressor inlet and outlet temperature, as:

$$W_C = \eta_C^{-1}\dot{N}\hat{C}_{p,1-2}T_1(\gamma^{\varphi} - 1) \tag{18}$$

3.3 Recuperator

The recuperator effectiveness α is defined as the ratio of the heat absorbed by the cold fluid in the regenerator, to the theoretical heat released by the hot fluid from inlet temperature to the inlet temperature of the cold fluid, written as:

$$\alpha = \frac{(1-\beta)(\hat{C}_{p,3}T_3 - \hat{C}_{p,2}T_2)}{\hat{C}_{p,5}T_5 - \hat{C}_{p,2}T_2} \tag{19}$$

Therefore, the temperature of reactor inlet T_3 can be written as:

$$T_3 = \frac{\alpha(\hat{C}_{p,5}T_5 - \hat{C}_{p,2}T_2) + (1+\beta)\hat{C}_{p,2}T_2}{(1-\beta)\hat{C}_{p,3}} \tag{20}$$

3.4 Bleed line

The heat created by the generator Q_G can be expressed as:

$$Q_G = (1-\eta_G)(W_T - W_C) \tag{21}$$

where η_G is generator efficiency. This heat is carried away by the bleed line:

$$Q_G = \beta\dot{N}\hat{C}_{p,2-8}(T_2 - T_8) \tag{22}$$

3.5 Reactor

The reactor thermal power is expressed in terms of the temperature of reactor inlet and outlet, as:

$$Q_R = (1-\beta)\dot{N}\hat{C}_{p,3-7}(T_7 - T_3) \tag{23}$$

3.6 Mixing box

According heat conservation theorem, one can get:

$$\hat{C}_{p,4}T_4 = (1-\beta)\hat{C}_{p,7}T_7 + \beta\hat{C}_{p,8}T_8 \tag{24}$$

3.7 Iterative solution

The cold side temperature T_1, and the hot side temperature T_7 are need to be given in advance. The iterative solution process for other temperature is: given initial value of T_4, and according to equation (12), (17), (20) and (22), T_5, T_2, T_3, and T_8 can be solved in order, and then the new value of T_4 can be obtained based on equation (24).

In the end, the net system efficiency of CBC is expressed as:

$$\eta = \frac{\eta_G(W_T - W_C)}{Q_R} \tag{25}$$

4. THERMODYNAMIC ANALYSIS

Adiabatic coefficient, recuperator effectiveness and normalized pressure loss of CBC cycle will change with different molar weight of helium-xenon mixture. The normalized pressure loss is defined as the ratio of the pressure loss in which the helium-xenon mixture is used, to that of pure helium, $f = (\Delta P / P)/(\Delta P / P)_{\mathrm{He}}$. ξ_1 and ξ_2 of CBC with helium are assumed to be constant, as $\xi_{1,\mathrm{He}}$ and $\xi_{2,\mathrm{He}}$ respectively.

Fig. 3 shows these three key parameters changing with molar weight increases. Adiabatic coefficient increases with the increase of the molar weight of helium-xenon mixture, from 0.400 to 0.414. Recuperator effectiveness firstly increases with molar weight increases, and then decreases, and the maximum value 0.81 is obtained in the vicinity of helium-8.6%xenon (15 g/mole). The recuperator effectiveness of the CBC where pure helium is used, is assumed to be 0.8. And the normalized pressure loss increases monotonically with the increase of molar weight.

The influence of these three key parameters change on the net efficiency of CBC is analyzed in below. During analysis, turbine adiabatic efficiency η_T is assumed to be 0.89; compressor adiabatic efficiency η_C is assumed to be 0.87; generator efficiency η_G is assumed to be 0.98; pressure loss of CBC where pure helium is used, $\xi_{1,\mathrm{He}}$ and $\xi_{2,\mathrm{He}}$, are assumed to be 0.015 and 0.009 respectively; bleed fraction β is assumed to be 0.02; and the cold side temperature T_1 and hot side temperature T_7 of are set to be 400 K and 1300 K.

Fig. 4 shows the influence of the change of adiabatic coefficient, recuperator effectiveness and normalized pressure loss, on the net efficiency of CBC. When analyzing the effect of one parameter on the cycle net efficiency, the other two parameters remain unchanged, as equal to that with pure helium (adiabatic coefficient is 0.400, recuperator effectiveness is 0.80, normalized pressure loss is 1). Fig. 4 illustrates that the change of adiabatic coefficient has relatively little effect on cycle net efficiency; cycle optimum net efficiency (corresponded to optimum compressor pressure ratio) increases with the increase of recuperator effectiveness; while cycle optimum net efficiency decreases with the increase of normalized pressure loss.

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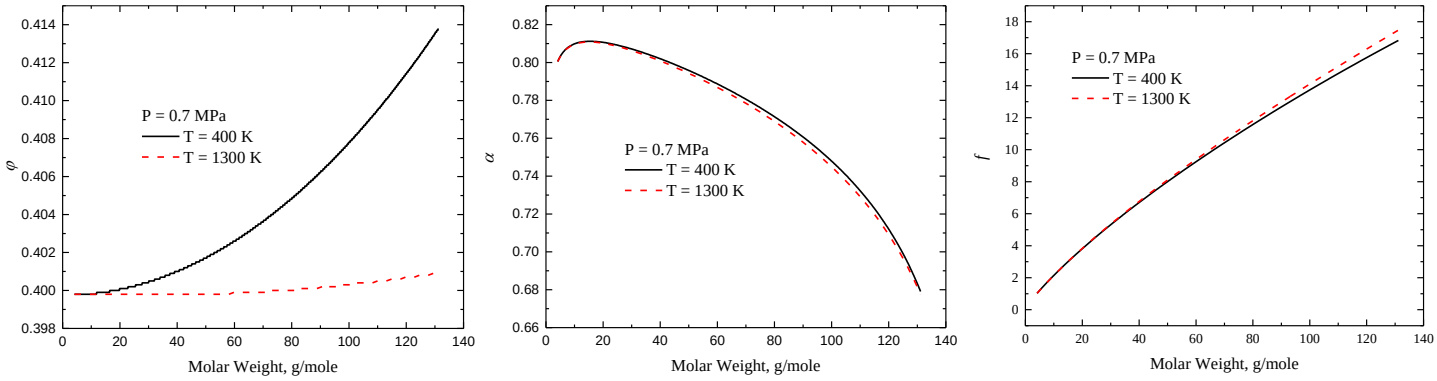


Fig 3. Key parameters of CBC affected by molar weight

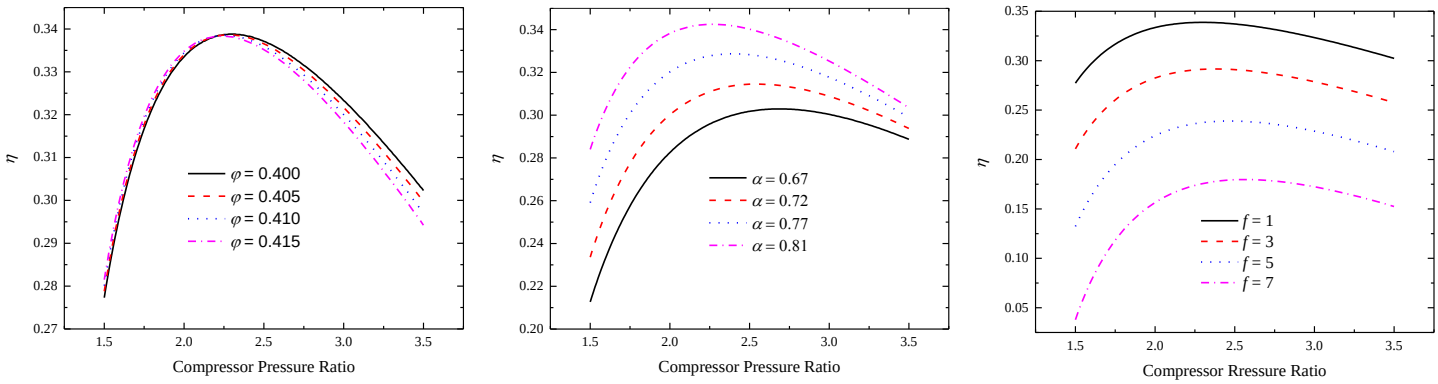


Fig. 4 Influence of three key parameters on net system efficiency

According to the previous analysis, with more xenon added in helium, there are both "positive effect" and "negative effect" on space closed Brayton cycle. Therefore, three different molar weight of helium-xenon mixtures are selected for comprehensive thermodynamic analysis. Working fluid 1: pure helium; working fluid 2: helium-8.6%xenon (corresponded to maximum recuperator effectiveness); working fluid 3: helium-29.7%xenon (has same recuperator effectiveness as pure helium), as shown in Fig. 5.

Thermodynamic analysis for CBC with above three working fluid was performed. System pressure P_2 was set to be 0.7 MPa, and cold / hot side temperature T_1 / T_7 were set to be 400 / 1300 K. The thermodynamic analysis results are shown in Fig. 5. Although pure helium leads to the highest net system efficiency, helium-8.6%xenon (working fluid 2) was adopted, for reducing series of turbine machinery to meet the demand of space craft[10], and net system efficiency of 29.18% was obtained in the end.

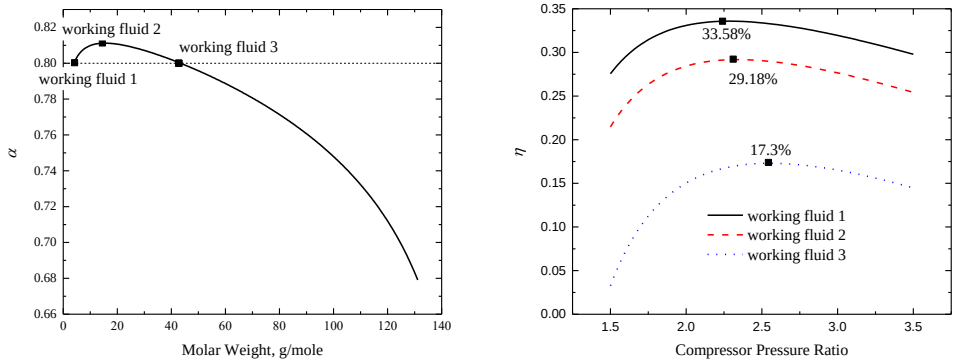


Fig. 5 Net System efficiency with different working fluid

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5. CONCLUSIONS

This paper introduced the semi-empirical calculations for the physical properties prediction of helium-xenon mixture. After that, the thermodynamic analysis model of space closed Brayton cycle was established. Then, the influences of molar weight of helium-xenon mixture on three key parameters (adiabatic coefficient, recuperator effectiveness, and normalized pressure loss) were performed, and their effects on net system efficiency were also carried out. Finally, helium-8.6%xenon was adopted as the working fluid and net system efficiency of 29.18% was obtained, according to the comparison of different molar weight of helium-xenon mixtures as working fluids.

NOMENCLATURE

B	second viral coefficient, mole/m^3
C	third viral coefficient, mole^2/m^6
CBC	closed Brayton cycle
C_p	specific heat capacity at constant pressure, J/(kg·K)
C_v	specific heat capacity at constant volume, J/(kg·K)
$\hat{C}_p$	molar specific heat capacity at constant pressure, J/(mole·K)
$\hat{C}_v$	molar specific heat capacity at constant volume, J/(mole·K)
M	molar weight, g/mole
M_1	molar weight of helium, g/mole
M_2	molar weight of xenon, g/mole
Q_G	heat created by generator, J
Q_R	reactor thermal power, J
P	pressure, Pa
R	perfect gas constant, R=8.31441 J/(mole·K)
$SNRP$	space nuclear reactor power
T	temperature, K
W_T	mechanical power developed in turbine, J
W_C	mechanical power used in compressor, J
x_1	molar fraction of helium
x_2	molar fraction of xenon
α	recuperator effectiveness
β	bleeding fraction
γ	compressor pressure ratio
η	net system efficiency
η_C	compressor adiabatic efficiency
η_G	generator efficiency
η_T	turbine adiabatic efficiency
λ	thermal conductivity, W / (m· K)
μ	Dynamic viscosity, Pa·s
π	turbine expansion ratio
ρ	density, kg/m^3
$\hat{\rho}$	molar density, mole/m^3
ξ	pressure loss ratio

ACKNOWLEDGMENTS

This work has been supported by the National Science and Technology Project (Grant No. ZX06901)

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